

Basics of CN

A computer network is a collection of interconnected devices that share resources and information. These devices can include computers, servers, printers, and other hardware. Networks allow for the efficient exchange of data, enabling various applications such as email, file sharing, and internet browsing.

Basic Terminologies

- **Network:** A group of connected computers and devices that can communicate and share data.
- **Node:** Any device that can send, receive, or forward data in a network. This includes laptops, mobiles, printers, earbuds, servers, etc.
- **Networking Devices:** Devices that manage and support networking functions. This includes routers, switches, hubs, and access points.
- **Transmission Media:** The physical or wireless medium through which data travels between devices.
- **Wired media:** Ethernet cables, optical fibre.
- **Wireless media:** Wi-Fi, Bluetooth, infrared

- **Service Provider Networks:** Networks offered by external providers that allow users or organisations to lease network access and capabilities. This includes internet providers, mobile carriers, etc.

Working Of a Computer Network

The basic building blocks of a Computer network are Nodes and Links.

- **Network Node:** Any device in a network, such as a modem, router, or computer, that sends or receives data.
- **Link:** The medium that connects nodes, like cables, wires, or wireless signals.
- **Working:** Networks use rules called protocols to send and receive data between devices. Each device is identified by a unique IP Address.
- **Firewall:** A security tool (hardware or software) that monitors traffic. Based on set rules, it either allows, blocks, or drops data.

Types of Computer Network Architecture

Computer Network falls under these broad Categories:

- **Client-Server Architecture:** Client-Server Architecture is a type of Computer Network Architecture in which Nodes can be Servers or Clients. Here, the server node can manage the Client Node Behaviour.
- **Peer-to-Peer Architecture:** In P2P (Peer-to-Peer) Architecture, there is not any concept of a Central Server. Each device is free for working as either client or server.

Types of Computer Networks

A computer network is a group of two or more independent computers and devices connected to share data, resources, and services. These connections can be established using wired media or wireless technologies.

Imagine sending a message from your smartphone to a friend's laptop or accessing files stored on a remote server. All of this is possible because of a computer network.

Types of Computer Networks are classified based on several factors:

1. Geographical area
2. Transmission medium

3. Ownership and access control

Classification Based on Geographical Area

Based on geographical area, computer networks are mainly divided into five types:

1. Personal Area Network (PAN)

A **Personal Area Network (PAN)** connects personal devices within a very short range around a single user.

- **Coverage Range:** 1–10 meters
- **Key Technologies:** Bluetooth, NFC, Infrared
- **Real-World Example:** Smartphone connected to wireless earbuds
- **When It Is Used:** For short-range communication between personal devices

2. Local Area Network (LAN)

A **Local Area Network (LAN)** connects computers and devices within a limited area such as a home, office, or building.

- **Coverage Range:** One room, building, or small campus
- **Key Technologies:** Ethernet, Wi-Fi
- **Real-World Example:** Home or office Wi-Fi network
- **When It Is Used:** For fast and secure communication within a local location

3. Campus Area Network (CAN)

A **Campus Area Network (CAN)** connects multiple LANs within a campus or a group of nearby buildings.

- **Coverage Range:** University or corporate campus
- **Key Technologies:** Ethernet, Fiber Optics
- **Real-World Example:** College or corporate campus network
- **When It Is Used:** To interconnect LANs across a campus under one organization

4. Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** connects multiple networks within a city or metropolitan region.

- **Coverage Range:** City or large town

- **Key Technologies:** Fiber Optics, Microwave, Metro Ethernet
- **Real-World Example:** City-wide ISP network
- **When It Is Used:** To provide high-speed connectivity across a city

5. Wide Area Network (WAN)

A **Wide Area Network (WAN)** connects networks over large geographical areas such as countries or continents.

- **Coverage Range:** Country, continent, or global
- **Key Technologies:** Leased lines, Satellite links, Internet
- **Real-World Example:** The Internet
- **When It Is Used:** To enable long-distance and global communication

Classification Based on Ownership and Access Control

Ownership and access control refer to who owns a network and who is permitted to use it. This classification focuses on administrative control and accessibility, rather than transmission medium, size, or technology.

Based on transmission technology, computer networks are mainly divided into three types:

1. Private Network

- Owned and managed by an individual, organization, or institution
- Access is restricted to authorized users only
- Requires authentication such as usernames, passwords, VPNs, or access tokens
- Used mainly for internal communication
- Offers higher security, privacy, and control
- Can be implemented using wired or wireless technologies
- Commonly used in companies, schools, banks, and government organizations

Examples:

- Corporate intranet
- Office LAN
- Private cloud network

2. Public Network

- Owned by service providers or government authorities
- Accessible to the general public
- Requires little or no authorization to connect
- Offers wide accessibility but lower security

- Usually shared by a large number of users
- Can operate over both wired and wireless media

Examples:

- Internet
- Public Wi-Fi networks
- Mobile cellular networks

3. Hybrid Network

- Combination of private and public networks
- Some parts of the network are restricted, while others are publicly accessible
- Allows organizations to maintain internal security while providing external connectivity
- Common in modern enterprise and cloud-based systems

Examples:

- Corporate network connected to the Internet
- Enterprise cloud networks
- Online banking systems

Internetwork

- An internetwork is a collection of two or more independent networks connected together to function as a single logical network.
- It enables communication between devices across different networks.
- The Internet itself is the largest example of internetwork.

Key Characteristics

- Consists of multiple networks, not just individual computers
- Networks may differ in:
 - Size (LAN, MAN, WAN)
 - Technology (wired or wireless)
 - Architecture
- Uses networking devices to interconnect networks
- Communication is enabled through standard protocols

There are two main types, Intranet and Extranet.

1. Intranet

An intranet is a private network used within an organization. It is accessible only to authorized users such as employees.

Key Features

- Owned and controlled by a single organization
- Access restricted using login credentials
- Used for internal communication and collaboration
- Improves information sharing within the organization
- Uses Internet protocols like TCP/IP, HTTP, and FTP
- Can be implemented using wired or wireless networks

Uses

- Employee portals
- Internal emails and messaging
- Sharing company policies and documents
- Internal applications (HR, payroll, project management)

Examples

- Corporate internal website
- University campus portal
- Government department internal systems

2. Extranet

An extranet is an extension of an intranet that allows limited access to external users. it connects the organization's internal network with trusted outsiders.

Key Features

- Partially private network
- Access provided to partners, vendors, suppliers, or customers
- Requires authentication and authorization
- Improves business collaboration
- Often implemented using VPN or secure web access

Uses

- Business-to-business (B2B) communication
- Supply chain management
- Customer portals
- Partner collaboration platforms

Examples

- Supplier management system
- Online banking portals

- Vendor access portals